

PE-EC701A — Microwave Theory and Technique

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Module 4 Exam Notes: Microwave Network Analysis

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Equivalent Voltages and Currents for Non-TEM Lines

In waveguides supporting TE or TM modes, the physical voltage and current are not uniquely defined because the electric and magnetic fields vary spatially across the guide cross-section.

Equivalent Voltage and Current: Normalized wave quantities defined such that their ratio equals the wave impedance (Z_w) and their product represents the power flow (P) down the waveguide. They allow waveguides to be analyzed using standard transmission line theory.

Let E_t and H_t be the transverse electric and magnetic fields. The equivalent voltage $V(z)$ and current $I(z)$ are defined as:

$$V(z) = K_V \iint_S E_t \cdot e_n dS \quad (1)$$

$$I(z) = K_I \iint_S H_t \cdot h_n dS \quad (2)$$

Where e_n, h_n are transverse vector mode functions. The normalization is chosen such that:

$$P = \frac{1}{2} \text{Re}(VI^*) \quad \text{and} \quad \frac{V}{I} = Z_w \quad (3)$$

2. Scattering (S) Parameters

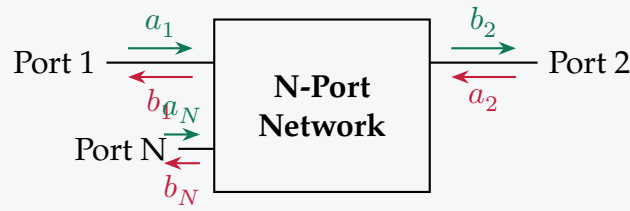
At high frequencies (microwave range), traditional low-frequency network parameters such as Impedance (Z), Admittance (Y), and Transmission ($ABCD$) parameters become extremely difficult to measure.

2.1 Why S-Parameters are Preferred at Microwaves

- **No Open/Short Circuit Requirements:** Measuring Z and Y parameters requires establishing perfect open and short circuits at the device terminals. At GHz frequencies, lead inductance and stray capacitance make perfect open/shorts impossible to realize.
- **Device Protection:** Creating a short circuit on active microwave devices (like Gunn diodes or high-frequency transistors) can cause them to become unstable and self-oscillate, potentially damaging the device.
- **Matched Terminations:** S-parameters are measured using **matched loads** (typically 50Ω). Matched loads are easy to construct at high frequencies and prevent signal reflections.
- **Direct Link to Wave Quantities:** S-parameters relate directly to incident and reflected power waves, which are easily measured using directional couplers.

2.2 S-Parameter Mathematical Definition

Consider an N -port microwave network. At each port i , we define normalized incident and reflected wave variables a_i and b_i :

N-Port Microwave Network Wave Variables**Scattering Parameter Relations**

The wave variables a_i (incident) and b_i (reflected) are related to port voltages V_i , currents I_i , and characteristic impedance Z_0 by:

$$a_i = \frac{V_i + I_i Z_0}{2\sqrt{Z_0}} \quad (\text{incident wave}) \quad (4)$$

$$b_i = \frac{V_i - I_i Z_0}{2\sqrt{Z_0}} \quad (\text{reflected wave}) \quad (5)$$

The linear relationship between the waves is expressed in matrix form as:

$$[b] = [S][a] \quad (6)$$

For a 2-port network:

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \quad (7)$$

Where:

$S_{ii} = \left. \frac{b_i}{a_i} \right|_{a_k=0 \text{ for } k \neq i}$ is the reflection coefficient at port i when all other ports are matched.

$S_{ij} = \left. \frac{b_i}{a_j} \right|_{a_k=0 \text{ for } k \neq j}$ is the transmission coefficient from port j to port i under matched termination.

3. Properties of the S-Matrix

The S-matrix possesses key mathematical properties governed by reciprocity and conservation of energy.

3.1 Symmetric Property for Reciprocal Networks

A network is reciprocal if the transmission of energy from port i to port j is identical to transmission from j to i .

Symmetric Property: The scattering matrix of a reciprocal network is symmetric.

$$[S]^T = [S] \implies S_{ij} = S_{ji} \quad (8)$$

3.1.1 Derivation of the Reciprocal Property

For a reciprocal network, the impedance matrix $[Z]$ is symmetric:

$$[Z]^T = [Z]$$

The relationship between S-parameters and Z-parameters (normalized to $Z_0 = 1$) is:

$$[S] = ([Z] - [I])([Z] + [I])^{-1} \quad (9)$$

Taking the transpose of both sides:

$$\begin{aligned} [S]^T &= ([Z] - [I])([Z] + [I])^{-1})^T \\ &= (([Z] + [I])^{-1})^T \cdot ([Z] - [I])^T \\ &= (([Z] + [I])^T)^{-1} \cdot ([Z]^T - [I]^T) \end{aligned}$$

Since $[Z]^T = [Z]$ and the identity matrix is symmetric ($[I]^T = [I]$):

$$[S]^T = ([Z] + [I])^{-1}([Z] - [I])$$

Since matrices of the form $([Z] \pm [I])$ commute:

$$([Z] + [I])^{-1}([Z] - [I]) = ([Z] - [I])([Z] + [I])^{-1} = [S]$$

Thus, we prove:

$$[S]^T = [S] \quad (\text{Q.E.D.}) \quad (10)$$

3.2 Unitary Property for Lossless Networks

In a lossless network, no real power is dissipated; the total power entering the network must equal the total power leaving it.

Unitary Property: The S-matrix of a lossless network is unitary.

$$[S]^{*,T}[S] = [I] \quad \text{or} \quad [S]^\dagger[S] = [I] \quad (11)$$

3.2.1 Derivation of the Lossless Property

The total power entering an N -port network is the sum of incident powers:

$$P_{in} = \frac{1}{2} \sum_{i=1}^N |a_i|^2 = \frac{1}{2} [a]^{*,T}[a]$$

The total power leaving the network is the sum of reflected powers:

$$P_{out} = \frac{1}{2} \sum_{i=1}^N |b_i|^2 = \frac{1}{2} [b]^{*,T}[b]$$

Since the network is lossless, conservation of energy dictates $P_{in} = P_{out}$:

$$[a]^{*,T}[a] = [b]^{*,T}[b]$$

Substituting $[b] = [S][a]$ and taking its conjugate transpose $[b]^{*,T} = [a]^{*,T}[S]^{*,T}$:

$$[a]^{*,T}[a] = [a]^{*,T}[S]^{*,T}[S][a]$$

This equation must hold for any arbitrary incident wave vector $[a]$. Hence:

$$[S]^{*,T}[S] = [I] \quad (\text{Q.E.D.}) \quad (12)$$

Quick Revision — Module IV

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Key Formulas at a Glance

Incident wave variable	$a_i = \frac{V_i + I_i Z_0}{2\sqrt{Z_0}}$
Reflected wave variable	$b_i = \frac{V_i - I_i Z_0}{2\sqrt{Z_0}}$
Reciprocal condition	$[S]^T = [S] \implies S_{ij} = S_{ji}$
Lossless condition	$[S]^{*,T}[S] = [I] \implies \sum_{k=1}^N S_{ki}^* S_{kj} = \delta_{ij}$
Reflection & VSWR	$S_{11} = \Gamma = \frac{\text{VSWR} - 1}{\text{VSWR} + 1}$

One-Line Definitions

- **S-Parameters:** Network parameters relating incident and reflected voltage waves.
- **Reciprocal Network:** A passive network where transmission coefficients are independent of propagation direction.
- **Lossless Network:** A network that conserves RF power, having zero internal power dissipation.
- **Unitary Matrix:** A complex matrix whose conjugate transpose is also its inverse.
- **Insertion Loss:** The attenuation of a signal passing through a network, given by $-20 \log_{10} |S_{ji}|$ dB.
- **Return Loss:** A measure of reflection mismatch at a port, given by $-20 \log_{10} |S_{ii}|$ dB.
- **Matched Termination:** Connecting a load equal to the characteristic impedance, preventing reflection.

Frequently Asked / Viva Questions

Q1. Why are Z and Y parameters not suitable for microwave frequencies?

Because they require open and short circuit terminations. At microwave frequencies, parasitics make open/shorts impossible, and shorts can damage active devices.

Q2. What is the physical meaning of S_{11} and S_{21} ?

S_{11} is the input reflection coefficient when port 2 is matched. S_{21} is the forward transmission coefficient from port 1 to port 2 when port 2 is matched.

Q3. State the unitary property of the S-matrix in summation form.

For a lossless network, $\sum_{k=1}^N |S_{ki}|^2 = 1$ (unity relation) and $\sum_{k=1}^N S_{ki}^* S_{kj} = 0$ for $i \neq j$ (zero relation).

Q4. Prove that a 1-port network with a matched load has $S_{11} = 0$.

Frequently Asked / Viva Questions (contd.)

A matched load absorbs all incident energy, generating zero reflected wave ($b_1 = 0$), so $S_{11} = b_1/a_1 = 0$.

Q5. What is the relationship between S-parameters and ABCD parameters?

ABCD parameters relate input voltage and current to output voltage and current. They are cascading parameters, unlike S-parameters which relate incident/reflected waves.

Q6. Why are S-parameters measured relative to a specific reference impedance (usually 50 Ω)?

Because wave variables a and b are normalized to $\sqrt{Z_0}$. Changing Z_0 changes the values of the S-parameters.

Q7. What does a return loss of ∞ dB represent?

It represents a perfect match ($S_{11} = 0$), where no signal is reflected back.

Q8. How do you check if an S-matrix represents a physical, passive network?

The magnitude of all elements must be less than or equal to one ($|S_{ij}| \leq 1$), and the network must not generate net power (the eigenvalues of $[S]^*{}^T[S]$ must be ≤ 1).

Q9. What is the relation between S-parameters and VSWR?

$$\text{VSWR} = \frac{1+|S_{ii}|}{1-|S_{ii}|}.$$

Q10. Under what conditions is the ABCD matrix preferred over the S-matrix?

When cascading multiple microwave networks in series, because the overall ABCD matrix is simply the product of individual ABCD matrices ($[ABCD]_{\text{total}} = [A] \cdot [B] \dots$).

Q11. What is equivocation in the context of network transmission?

It is the power lost or diverted due to reflection and mismatch at network junctions.

Q12. Why is the S-matrix symmetric for reciprocal networks?

Because the underlying electromagnetic materials are isotropic, yielding symmetric impedance ($[Z]$) and admittance ($[Y]$) matrices, which mathematically maps to $[S]^T = [S]$.

Comparison of Z, Y, ABCD, and S-Parameters

Parameters	Independent Variables	Dependent Variables	Terminations Required
Z Parameters	Currents (I_1, I_2)	Voltages (V_1, V_2)	Open Circuit ($I = 0$)
Y Parameters	Voltages (V_1, V_2)	Currents (I_1, I_2)	Short Circuit ($V = 0$)
ABCD Params	Output variables ($V_2, -I_2$)	Input variables (V_1, I_1)	Open / Short depending on parameter
S Parameters	Incident Waves (a_1, a_2)	Reflected Waves (b_1, b_2)	Matched Termination ($a = 0$)